

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (CL) Examination

May 2013

CL402 Computer Aided Structural Analysis

Date: 08.05.2013, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Write assumptions of plane strain problem.

[02]

(b) Do as directed:

[05]

- (i) Which are the types of elements are considered in FEM.
- (ii) Write a member flexibility matrix for a plane truss.
- (iii) Define: primary nodes.
- (iv) Define: secondary nodes.
- (v) Evaluate: $\int L_1^3 L_2^4 L_3^3 L_4^5 dV$.

Q - 2 Attempt any Two:

[14]

- (a) Using generalized co-ordinate approach, find the shape functions for two noded bar element shown in figure: 1.
- (b) Assemble the Jacobian Matrix corresponding to (0.57735, 0.57735) for element shown in figure: 2.
- (c) Derive the Shape Function for a one Dimensional body as shown in figure: 3 using Langrange functions. Draw the variations of shape function.

Q - 3 Attempt any two:

[14]

- (a) Analyze the frame using flexibility method member approach shown in figure: 4 and derive the B_{MS} and B_{RS} matrix. Take $EI = \text{constant}$. Considering the joint displacements are the rotation at A, B and at C.
- (b) Analyze the truss using Flexibility method member approach shown in figure: 5. All members are assumed to have the same axial rigidity EA . Determine end-actions, support reactions and joint displacements. $F_s = L/EA$ [4.828 0 -1 0 3.828 3.414; 0 1 0 0 0 -0.707; -1 0 1 0 -1 -0.707; 0 0 0 1 0 -0.707; 3.828 0 -1 0 3.828 2.707; 3.414 -0.707 -0.707 -0.707 2.707 4.828]
- (c) Analyze the two span continuous beam using flexibility method member approach shown in figure: 6. Derive the B_{MS} and B_{RS} matrix for the beam.

SECTION – II

- Q - 4 (a) Analyse the following beams using stiffness method shown in **figure: 7**. $EI =$ [03]
constant. Take $S^{-1} = 1/EI$ [2.256 -0.512 0.256; -0.512 1.025 -0.512; 0.256 -0.512
1.506]⁻
- (b) Analyze the propped cantilever beam shown in **figure: 8** by flexibility method system [04]
approach.
- Q - 5 (a) Analyse the truss and find the support reactions of truss as shown in **figure: 9** if, [14]
temperature of AB and AC is increased by 50° C and 25° C respectively. Take $AE =$
32000 kN; $\alpha = 0.00002$. Length of BC 3 mm more.

OR

- Q - 5 (a) Analyse the continuous beam shown in **figure: 10** using stiffness method member [14]
approach. Take EI as constant
- Q - 6 (a) Find Joint Displacement for grid shown in **figure: 11** using stiffness method member [14]
approach. Take $GJ = 0.8 EI$, $E = 2 \times 10^5 \text{ N/mm}^2$, $I = 0.001 \text{ mm}^4$

OR

- Q - 6 (a) Write the member stiffness matrix in member axes and rotation matrix for the space [06]
frame member.
- (b) Find the support reaction for the frame shown in **figure: 12** by stiffness method. Take [08]
 $A = 0.08 \text{ m}^2$ and $I = 1.067 \times 10^{-3}$.



Figure: 1

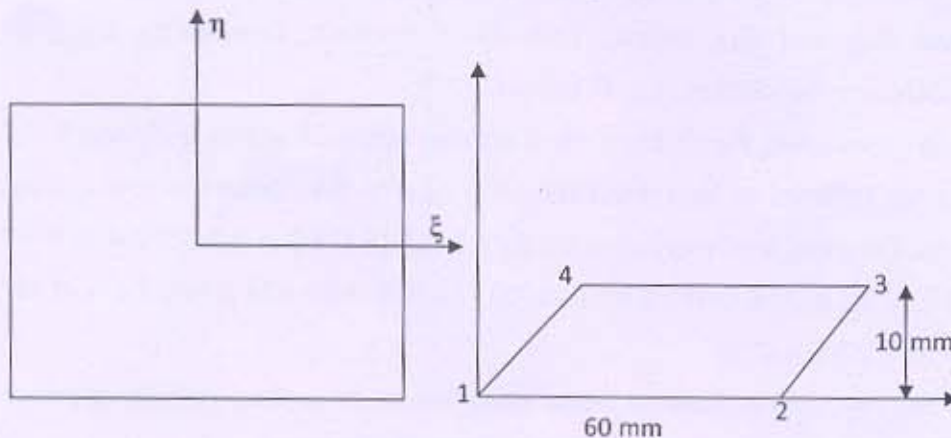


Figure: 2

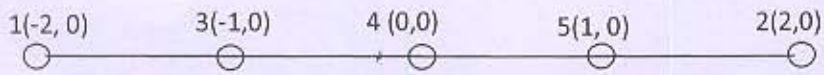


Figure: 3

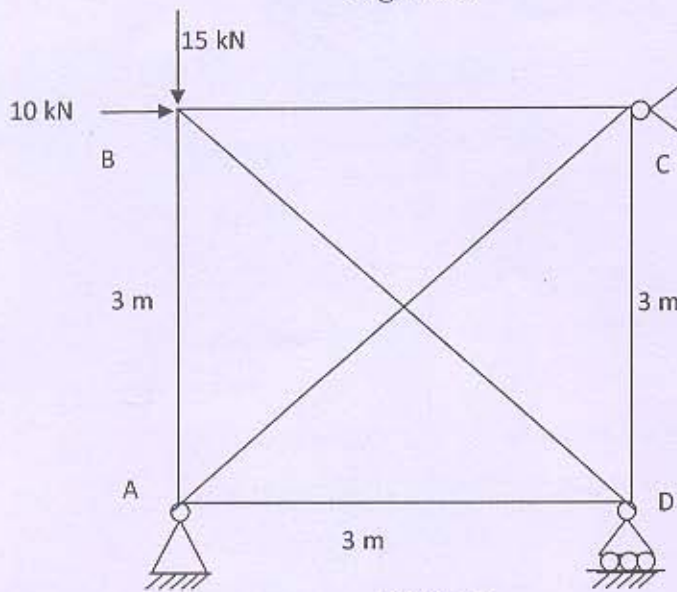


Figure: 5

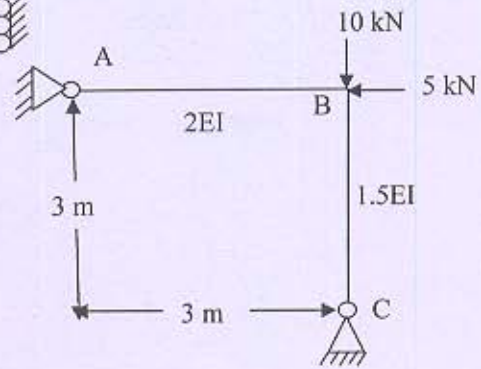


Figure: 4

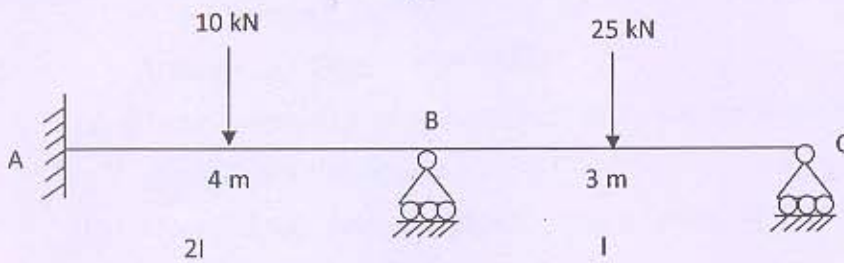


Figure: 6

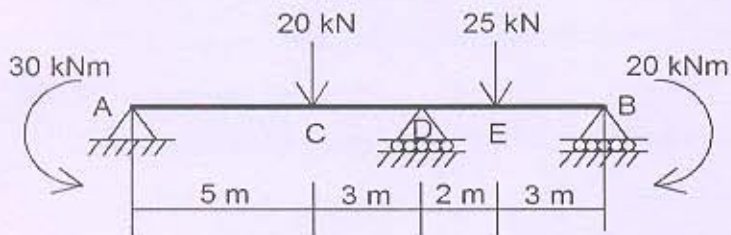


Figure: 7

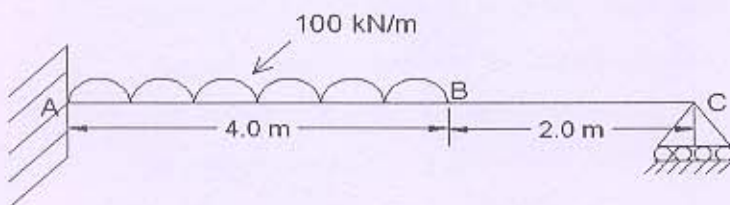


Figure: 8

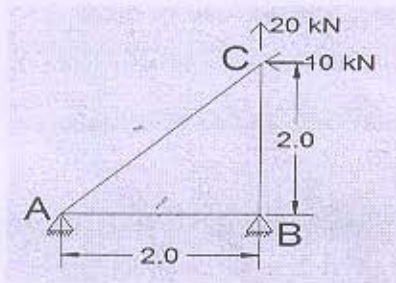


Figure: 9

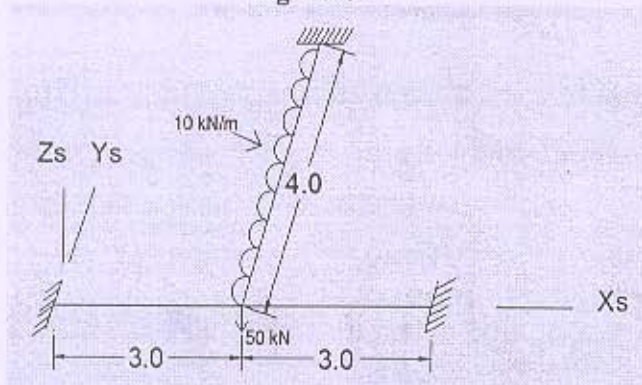


Figure: 11

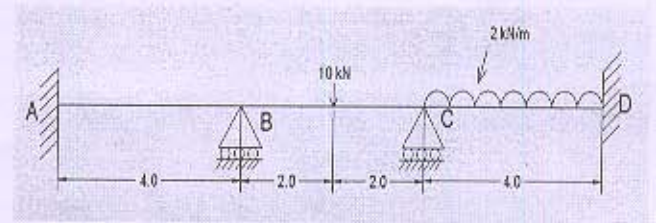


Figure: 10

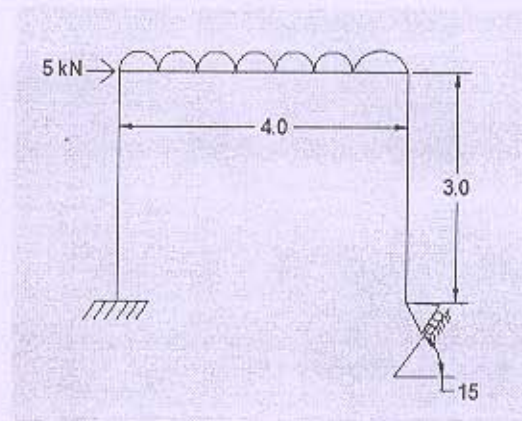


Figure: 12
